

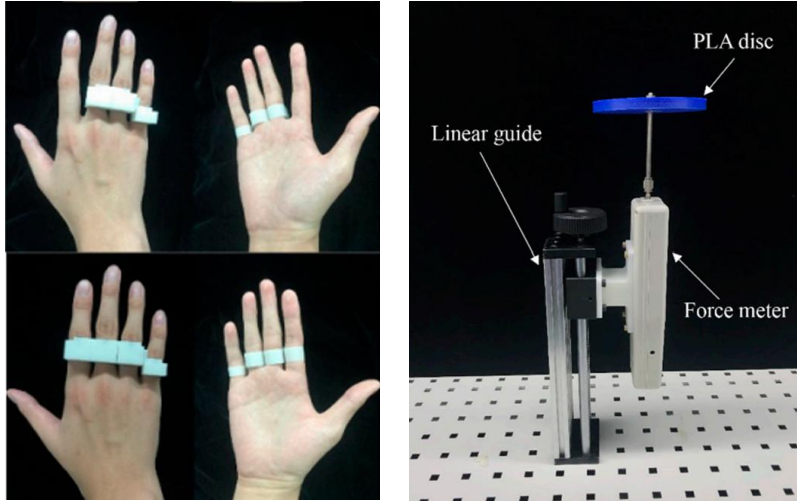
Enhancing Grasp Stability in Prosthetic Hands via Passive Finger Abduction/Adduction

Design of a Passive Abduction Mechanism Using Dual Torsion Springs,
Dual Tendon routing and Evaluation of Grasp Stability

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Background

Prior Study1 — Yan et al. (2021)¹



Grasp size $+4.67 \pm 1.93$ mm vs no-ABD

Grasp force $+5.27 \pm 4.25$ N vs no-ABD

Prior Study2 — Chang et al. (2021)²



Underactuated AB/AD DOF added at 3rd–5th finger MCP joints to prevent finger interference and enable diagonal grip

Both studies show that **AB/AD DOF** not only expands **grasp space/force** but also improves **contact-point (pressure) distribution** and **grasp stability**

→ **AB/AD is an essential DOF for stable grasping**

1. Yan, Y., Cheng, C., Guan, M., Zhang, J., & Wang, Y. (2021). The influence of the abduction joints of four fingers to grasp : Experimental and simulated verification. *Applied Sciences*, 11(24), 11960.
2. Chang, M. H., Kim, D. H., Kim, S., Lee, Y., Cho, S., Park, H., & Cho, K. (2021). Anthropomorphic prosthetic hand inspired by efficient swing mechanics for sports activities. *IEEE/ASME Transactions on Mechatronics*, 27(2), 1196–1207.

Background

Existing Prosthetic Hands

Prosthetics	ABD - None
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Ottobock-bebionic



Ossur - I limb



PSYONIC Ability Hand



Michelangelo

	Flexion/Extension only	+ ABD
Actuators	5~6	+ 4 (at least)
Weight	450g~500g	+ α

Extra DOF → Actuators, Size, Weights, Power Issues

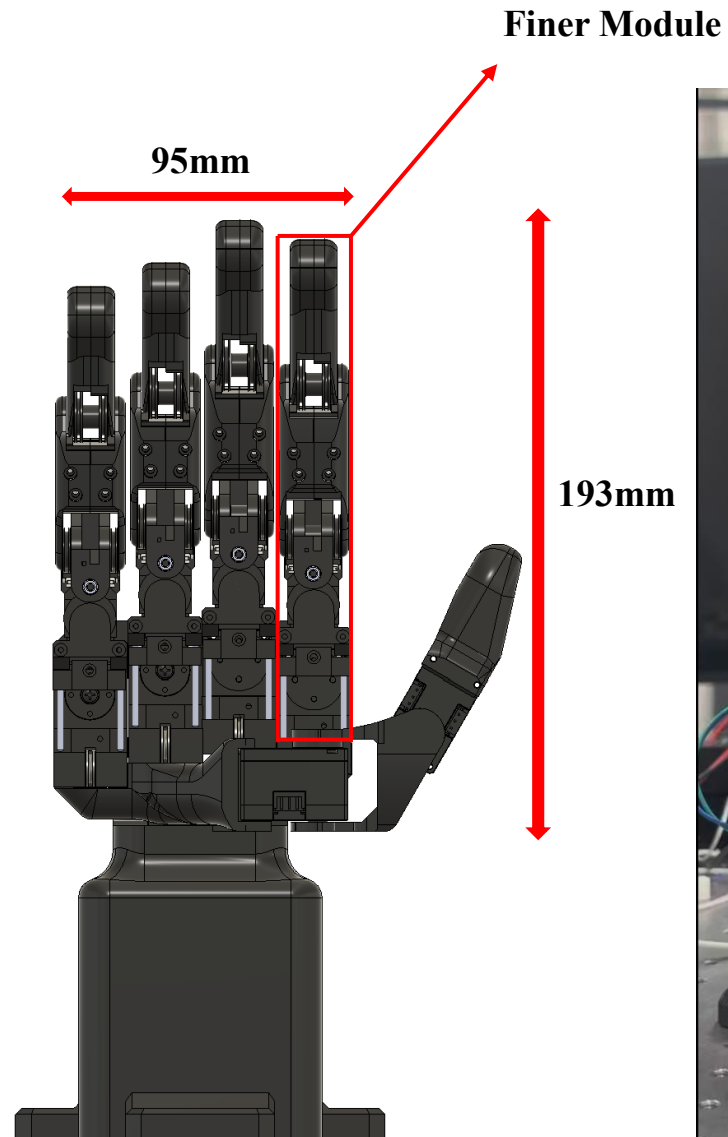
**GOAL 1: A mechanism achieving ab/ad DOF
without extra actuators**

GOAL 2: Achieve stable grasp of the proposed design

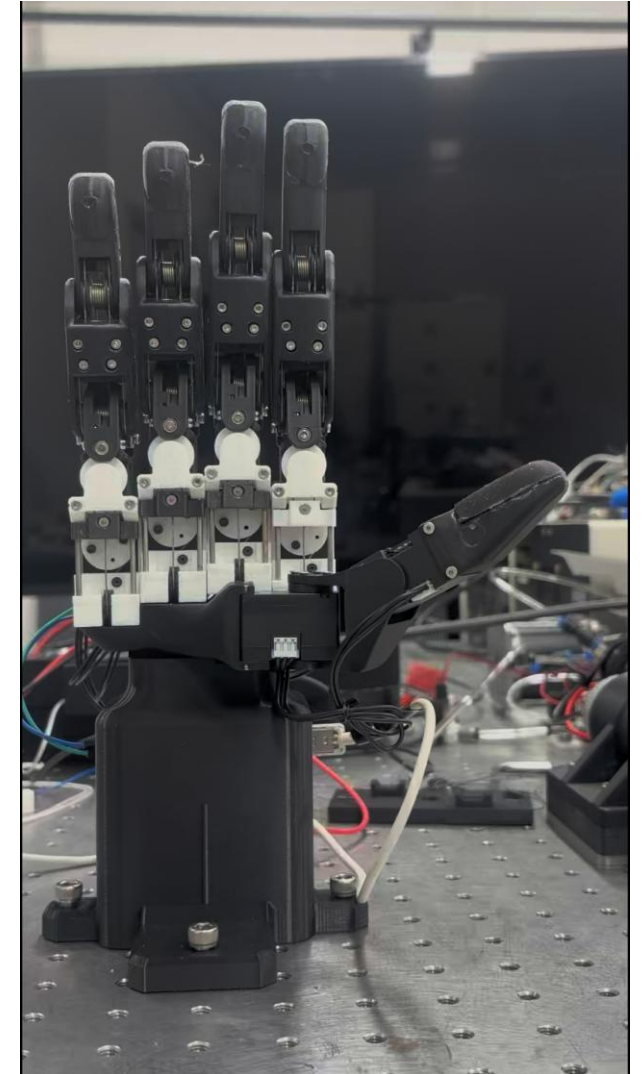
1. <https://www.ottobock.com/en-us/product/8E500-60921>
2. <https://www.ottobock.com/ko-kr/product/8E70-61081>
3. <https://www.ossur.com/ko-kr/%EB%B3%B4%EC%A1%B0%EA%B8%B0/%EC%95%94/i-limb-quantum>
4. <https://www.psyonic.io/ability-hand>

Hardware Design – Size & Spec

Weight	450 g
Hand length	193 mm
Hand width	95 mm
Actuators	6 x Dynamixel XC330-M288-T
Total DOF	14 (Thumb 2 + 4 fingers x 3)
Finger actuation	Tendon-driven
Finger ABD ROM	$\pm 25^\circ$
Thumb MCP ROM	160°
Finger MCP ROM	95°
Finger PIP ROM	85°
Thumb CMC ROM	115°

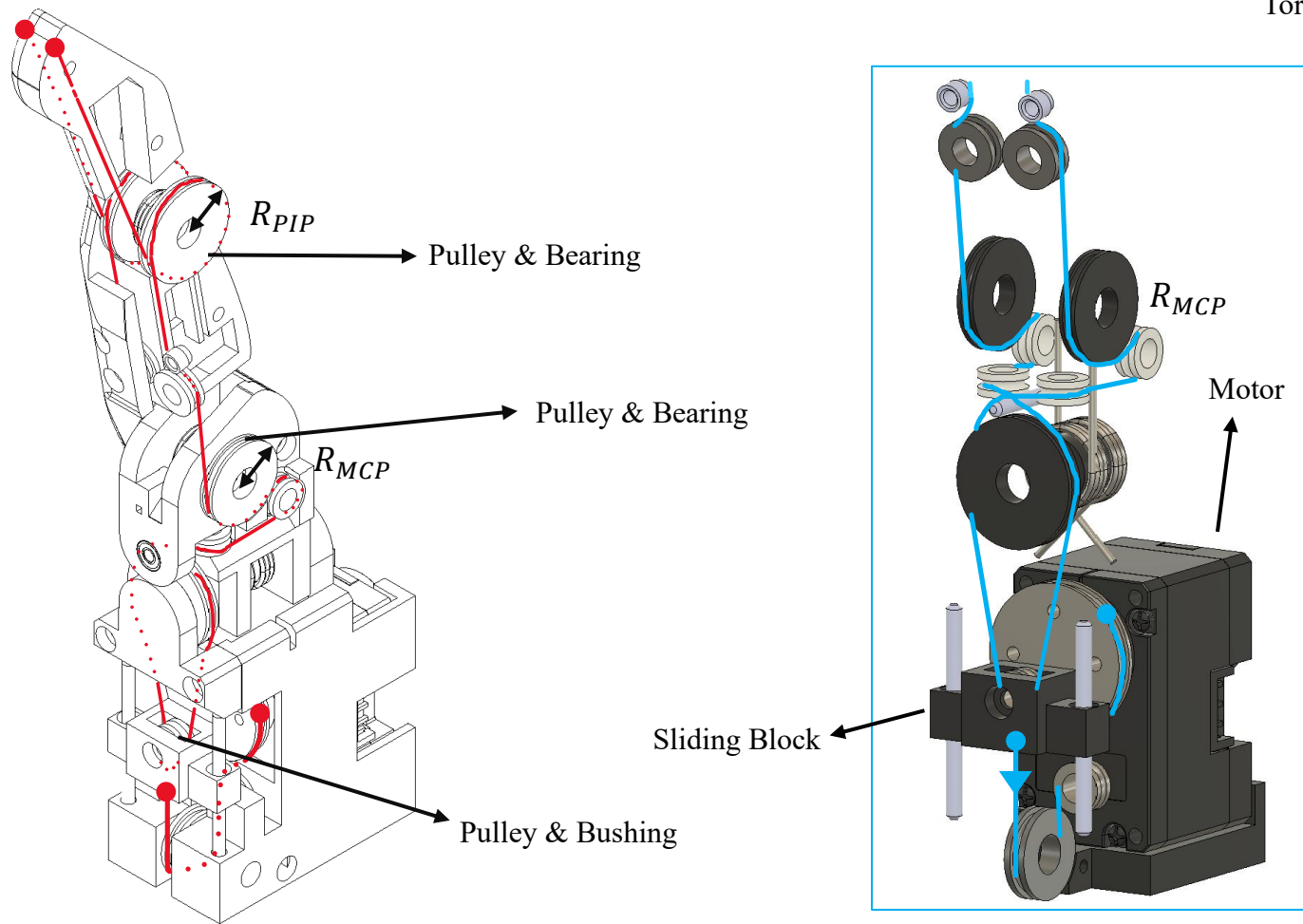


[CAD]



[Demo]

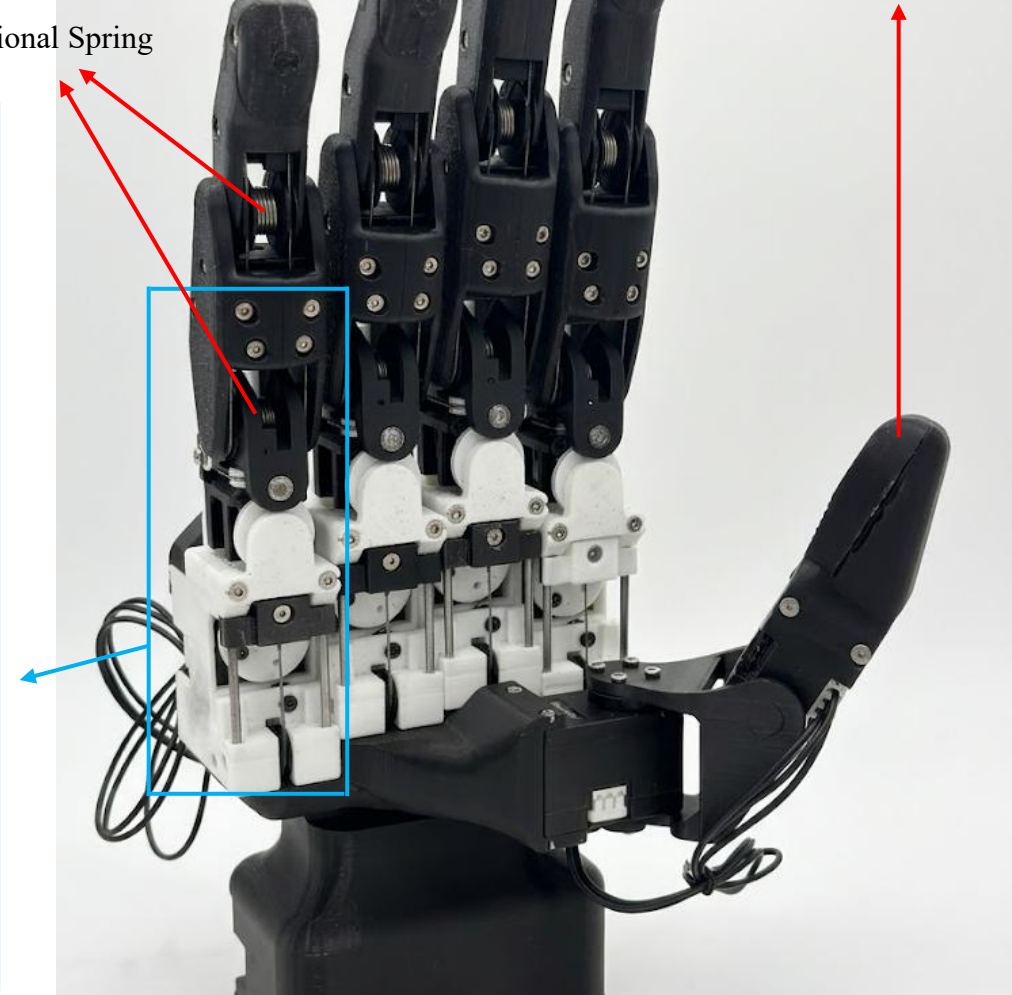
Hardware Design - Tendon Routing



[Tendon Routing & Finger Module]

Torsional Spring

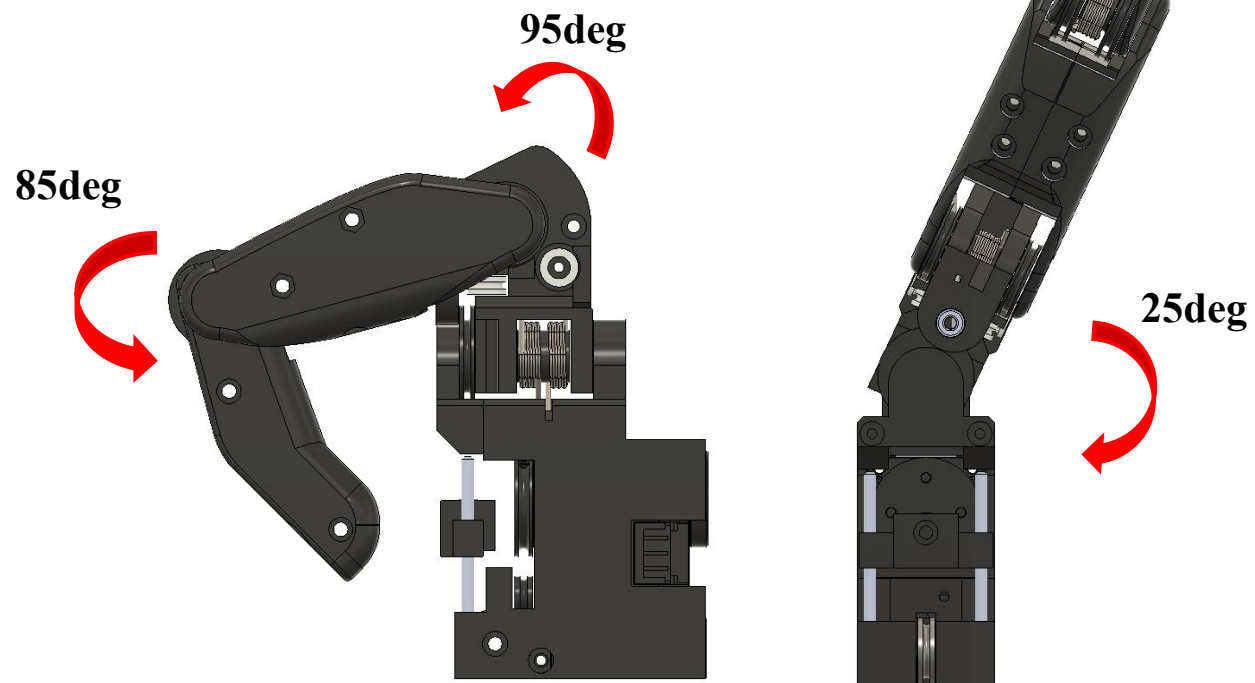
Silicon tips



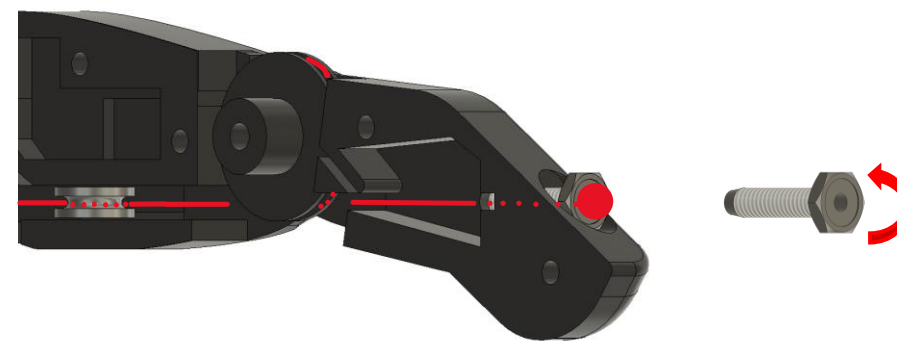
- Tendon routing designed to maintain a consistent tendon path throughout ab/ad motion

[Overview]

Hardware Design - Details

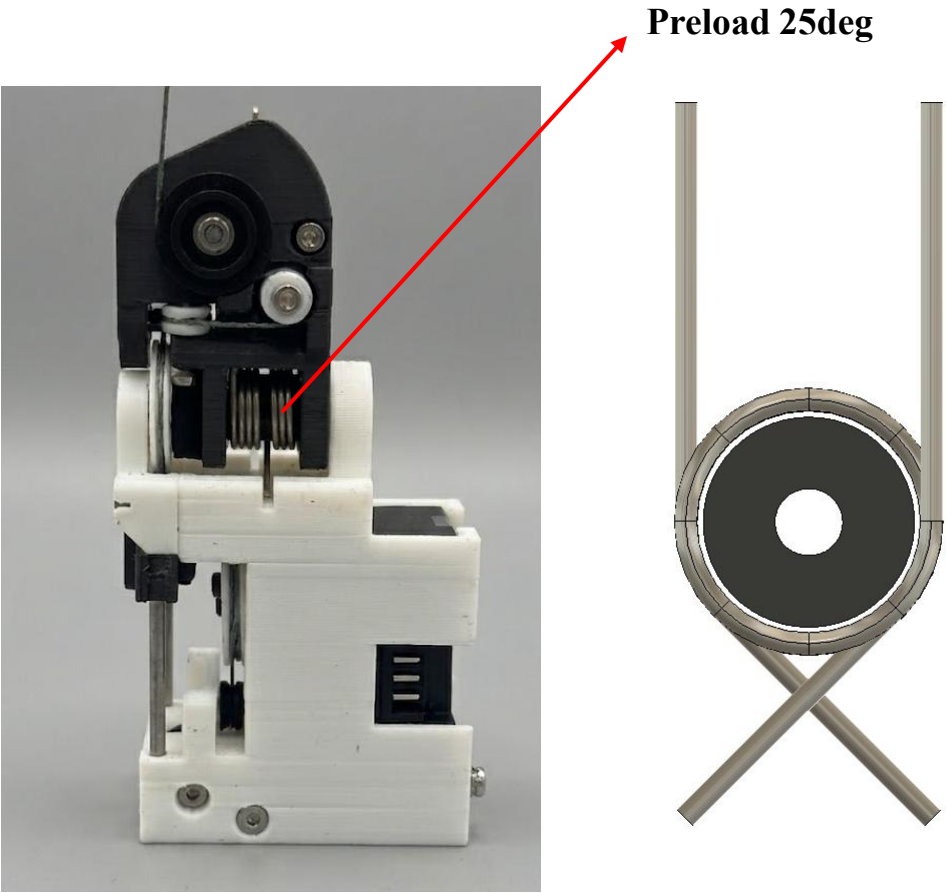


[ROM]



[Preventing Tendon Elongation]

Hardware Design – AB/AD



[Two Torsional Springs]

<i>$k(Nmm/deg)$</i>	<i>$2k\theta$</i>
0.656	1.312θ
1.08	2.16θ
1.58	3.16θ
2.01	4.02θ
3.30	6.6θ

$$\tau = k(25^\circ + \theta) - k(25^\circ - \theta) = 2k\theta$$

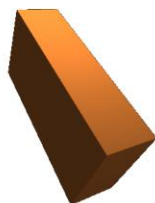
[Spring Candidates & Torque of ABD]

So how to choose stiffness k ?

- 1. Simulate various objects for each candidate stiffness k**
- 2. Identify the trend**
- 3. Select k based on the simulation results & equilibrium angle under gravity**
- 4. Validate the chosen k in a real-world environment**

Experiment - Simulation

Box
scale 1
150x75x40mm



Sphere
70mm



Ellipsoid
scale 0.9



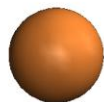
Disc
80mm



Box
scale 1.1
165x82x44mm



Sphere
80mm



Ellipsoid
scale 1



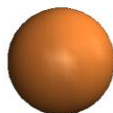
Disc
90mm



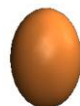
Box
scale 1.2
180x90x48mm



Sphere
90mm



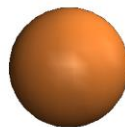
Ellipsoid
scale 1.1



Disc
100mm



Sphere
100mm



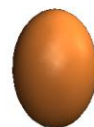
Ellipsoid
scale 1.2



Disc
110mm



Ellipsoid
scale 1.3



Disc
120mm



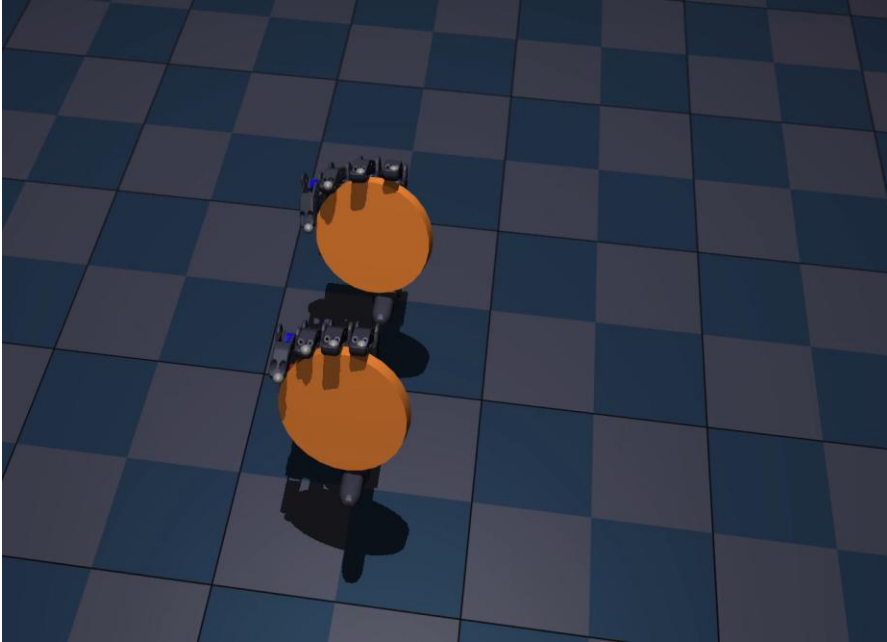
Disc
130mm



Object	Mass(g)	Trial
Sphere	30/50/70/90	$4 \times 4 = 16$
Disk	30/50/70/90	$6 \times 4 = 24$
Ellipsoid	30/50/70/90	$5 \times 4 = 20$
Box	80/100/120/140	$3 \times 4 = 12$

**Total 72 Objects $\times$ 8 trials
= 576 per k**

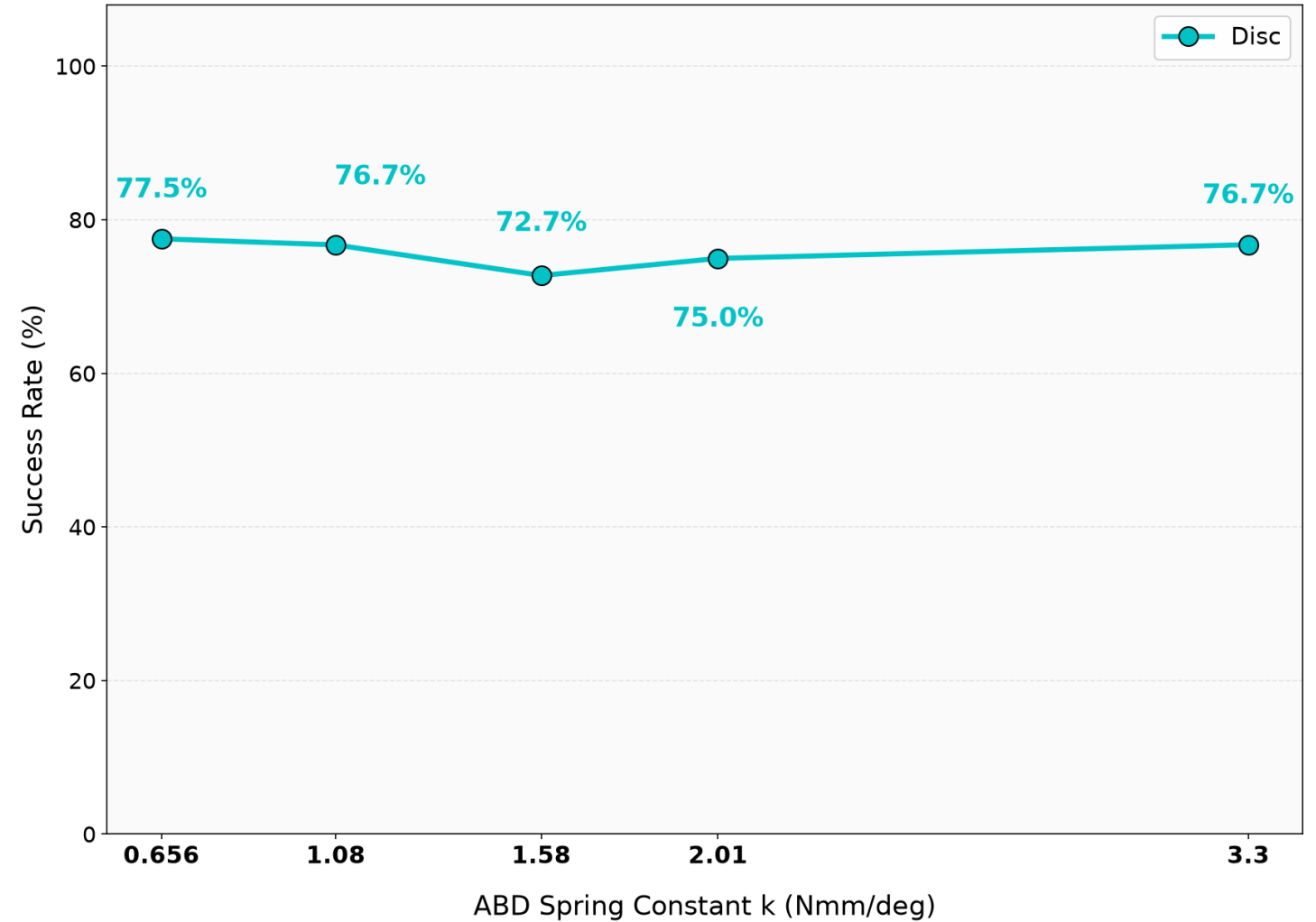
Experiment - Simulation



[Passive, Non passive Simulation]

Disk 120mm, 70g

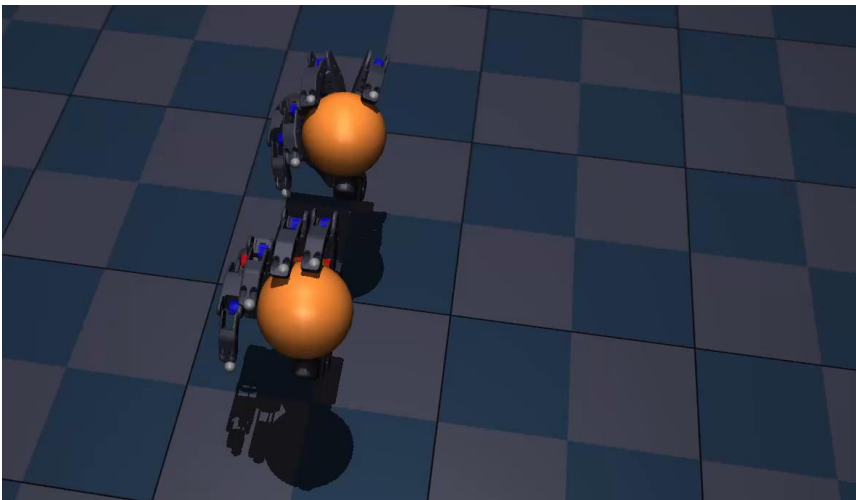
Disc — Success Rate vs ABD Spring Constant ($\tau=2k\theta$)



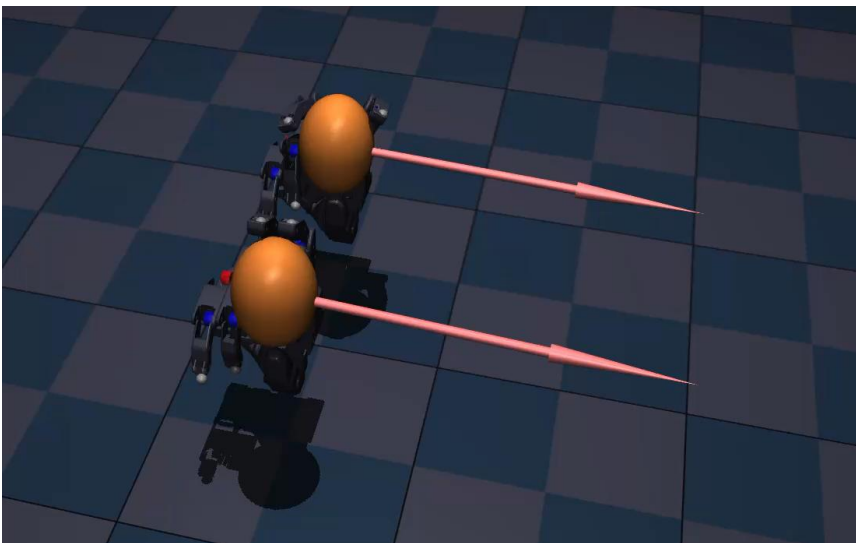
*All mass, size of Disk

*Success Rate: Passive win / Total trial

Experiment - Simulation

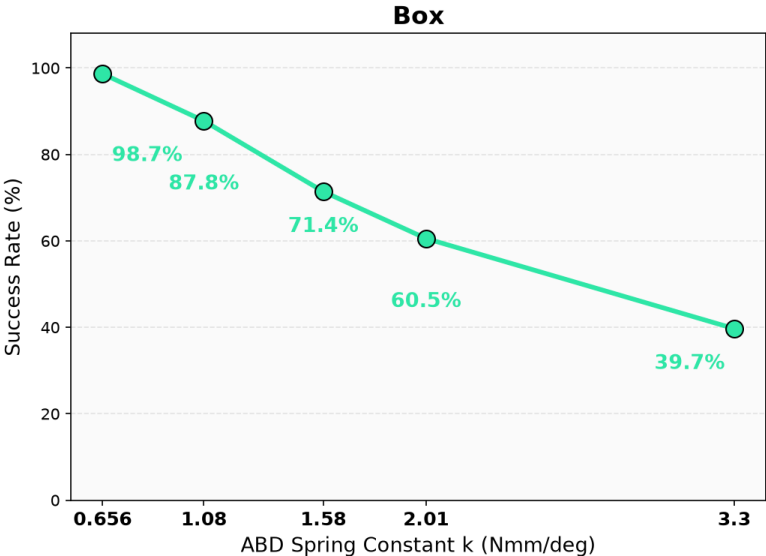
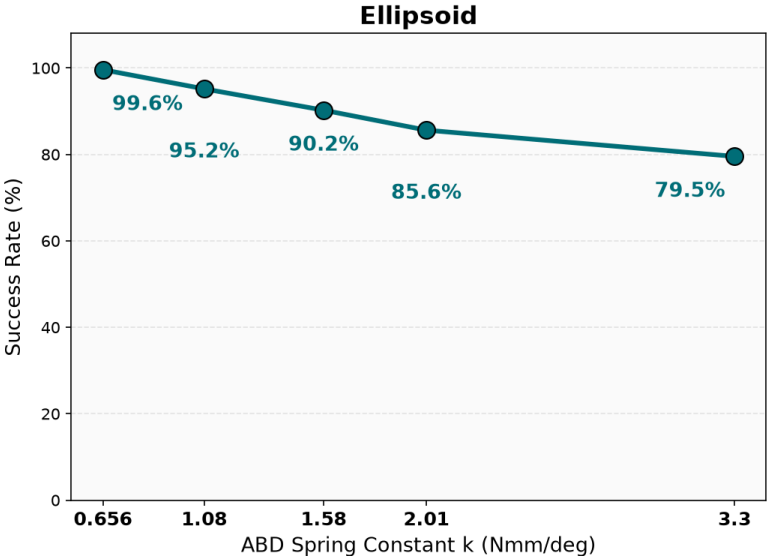
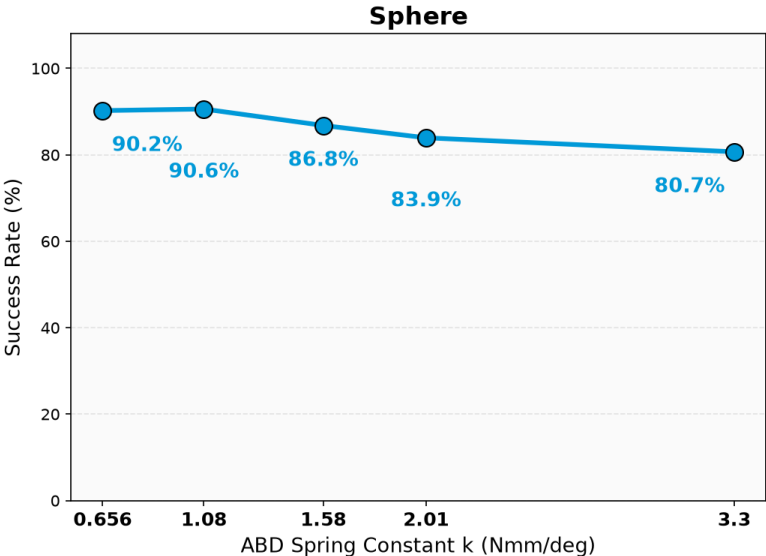
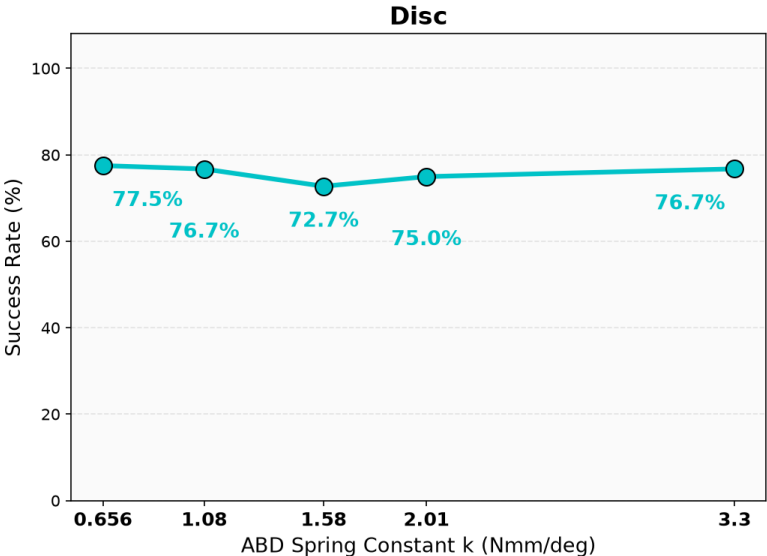


[Sphere 80mm, 70g]



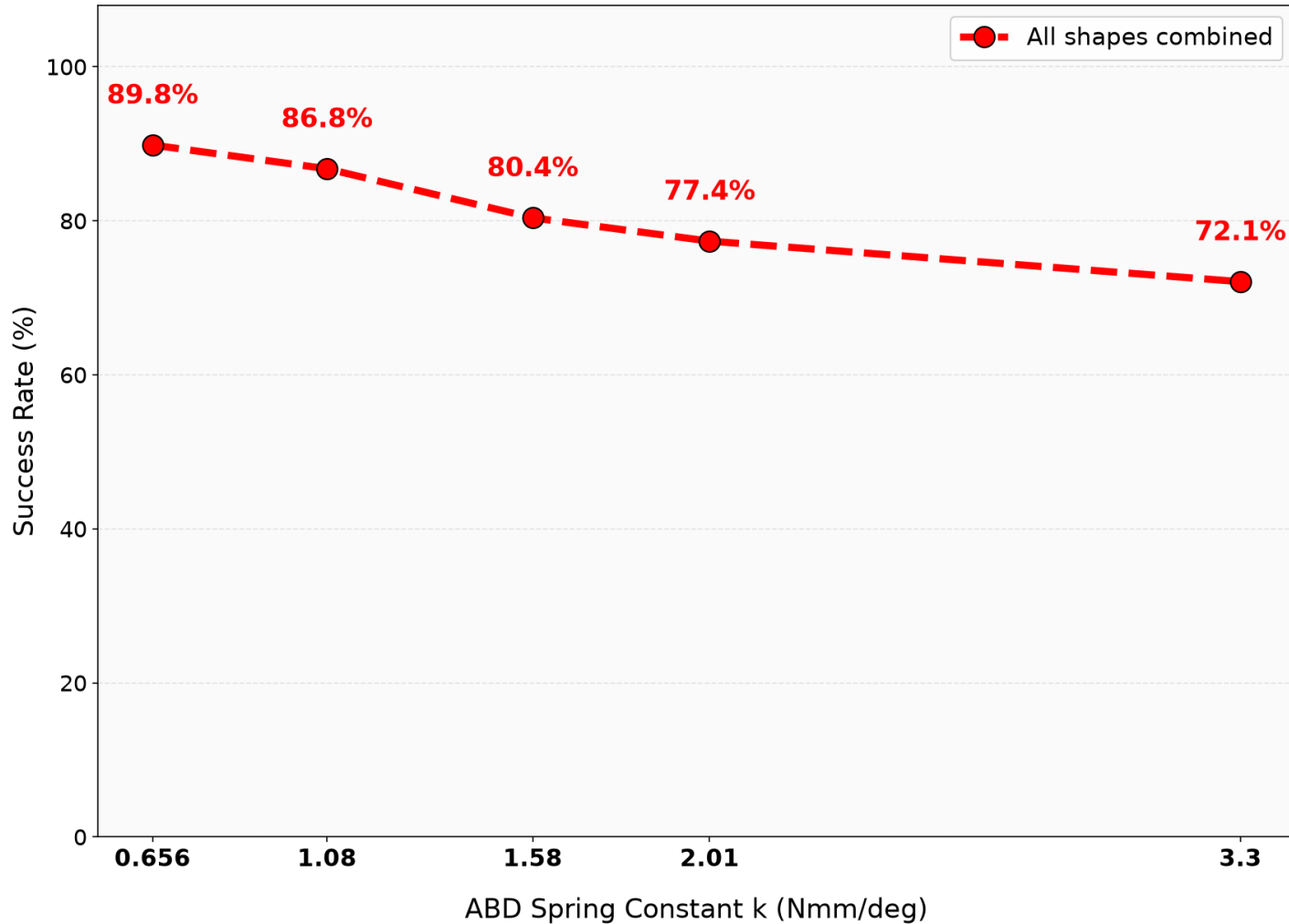
[Ellipsoid 70g]

Success Rate vs ABD Spring Constant ($\tau=2k\theta$), by shape



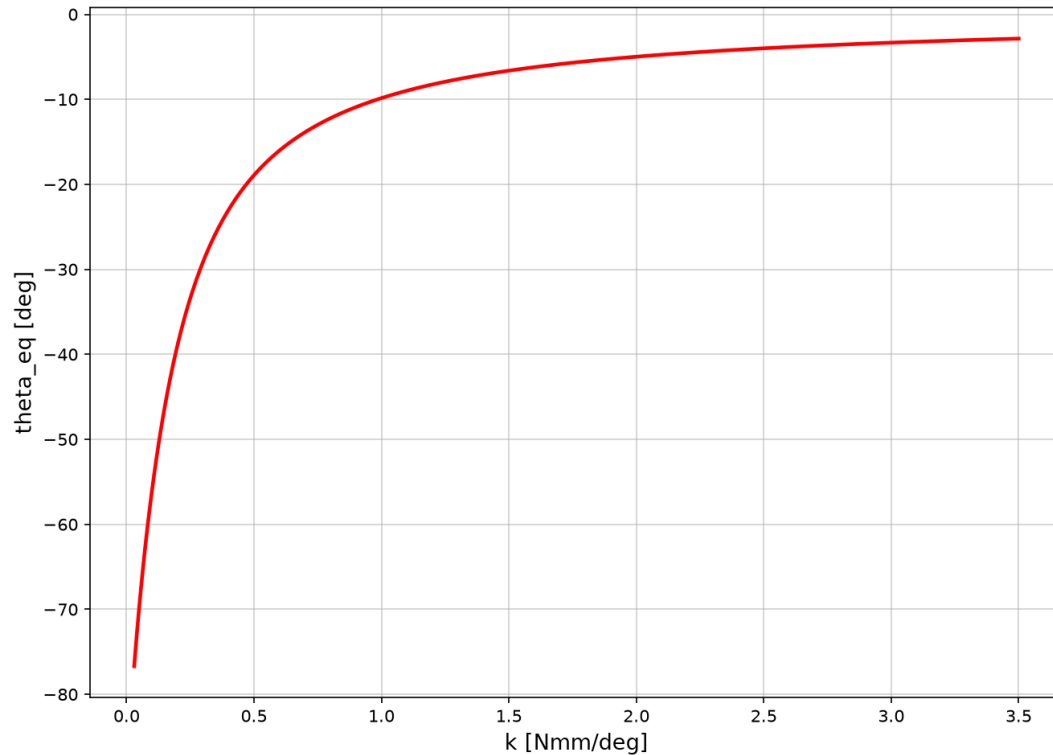
Experiment - Simulation

All Shapes Combined — Success Rate vs ABD Spring Constant ($\tau=2k\theta$)

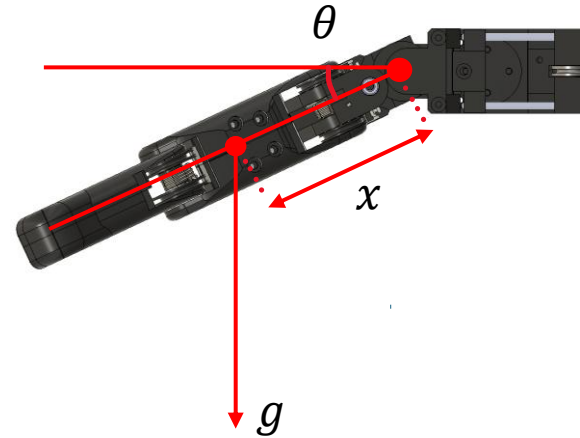


- Downward trend
- Lower stiffness k allows for a more adaptive passive AB/AD
- A more **adaptive grasp** achieves a **higher success rate**

Experiment - Simulation



[Equilibrium Angle under gravity θ_{eq} versus Stiffness k]



$$U(\theta) = \frac{1}{2}k(\theta - 25^\circ)^2 + \frac{1}{2}k(\theta + 25^\circ)^2$$

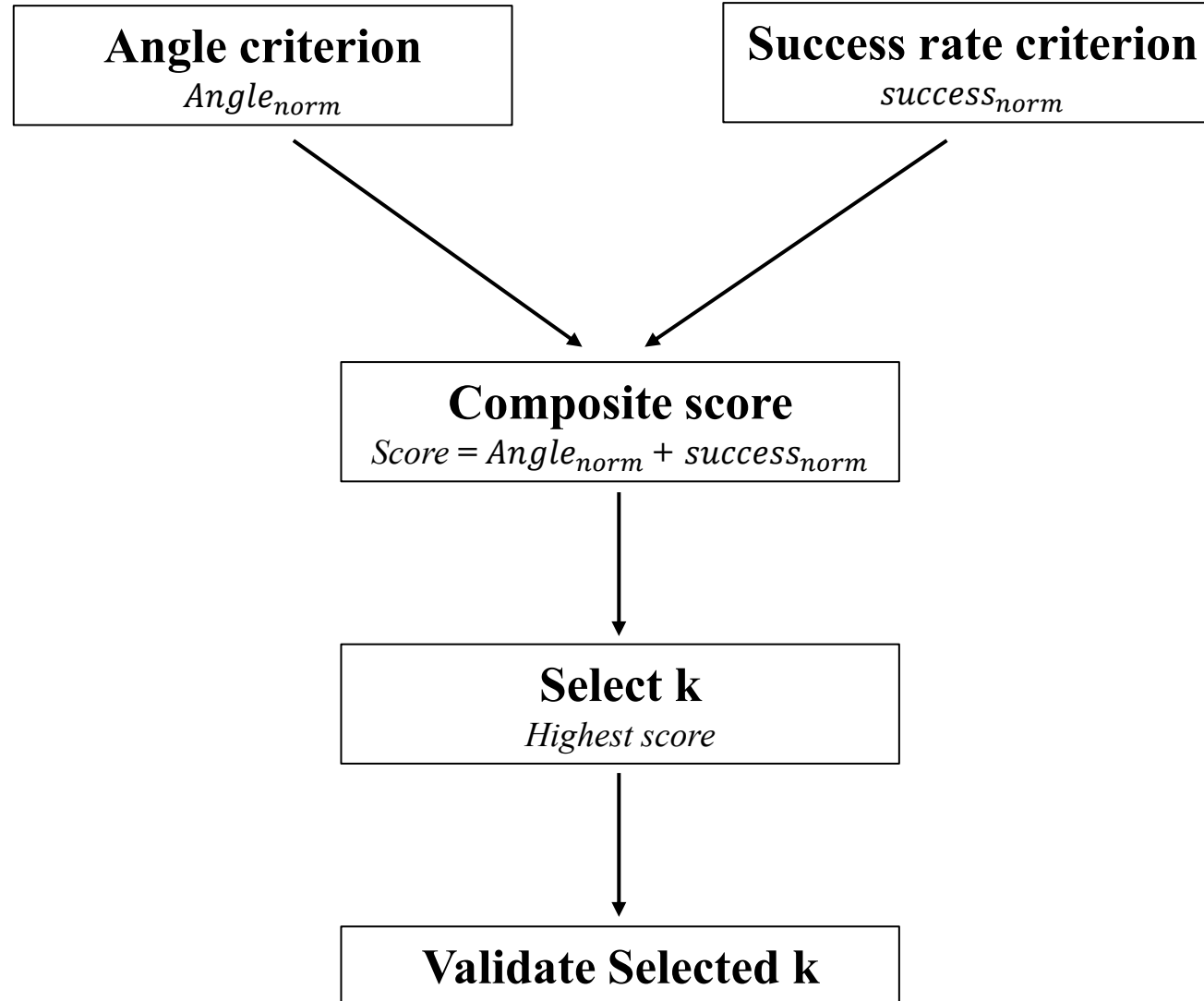
$$f(\theta) = U + mgx\sin\theta$$

$$\frac{df}{d\theta} = 0, \quad mgx = 0.02 \text{ Nm}$$

k	$\theta_{eq}(deg)$
0.656	-14.7
1.08	-9.1
1.58	-6.3
2.01	-5.0
3.30	-3.0

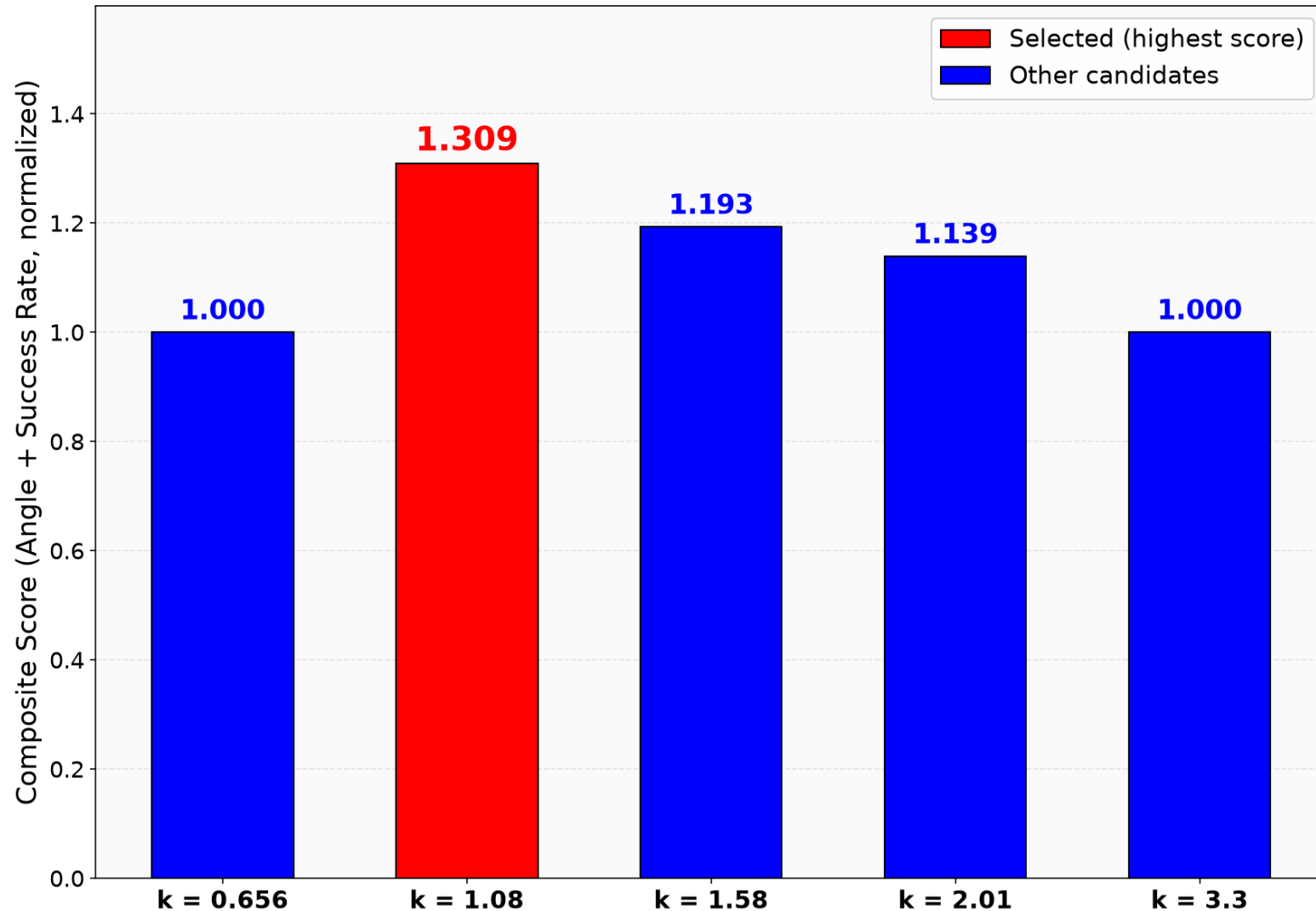
- Equilibrium angle offset θ_{eq} decreases as stiffness k increases

[Equilibrium Angle θ_{eq} according to Candidate k]



Experiment - Simulation

Composite Score by k — Higher Is Better



k	$ \theta_{eq} (deg)$	Success rate(%)
0.656	14.7	89.8
1.08	9.1	86.8
1.58	6.3	80.4
2.01	5.0	77.4
3.30	3.0	72.1

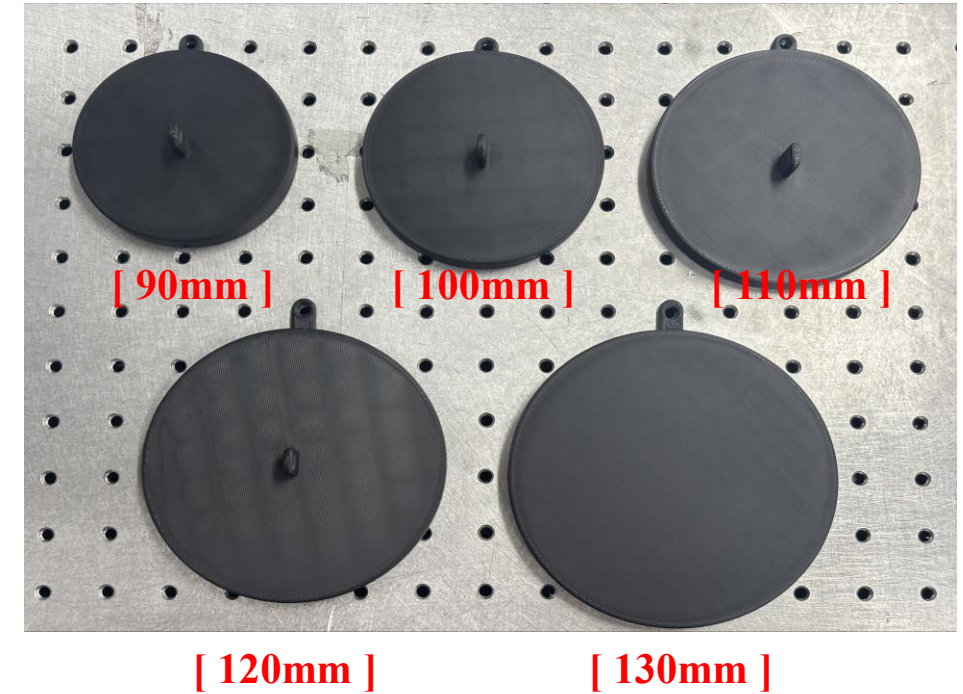
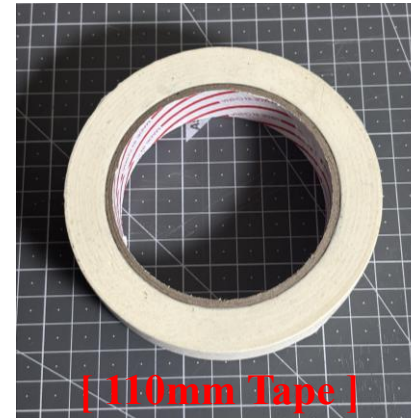
Min-Max Normalization

$$Angle_{norm} = \frac{\theta_{max} - |\theta_{eq}|}{\theta_{max} - \theta_{min}}$$

$$success_{norm} = \frac{success - success_{min}}{success_{max} - success_{min}}$$

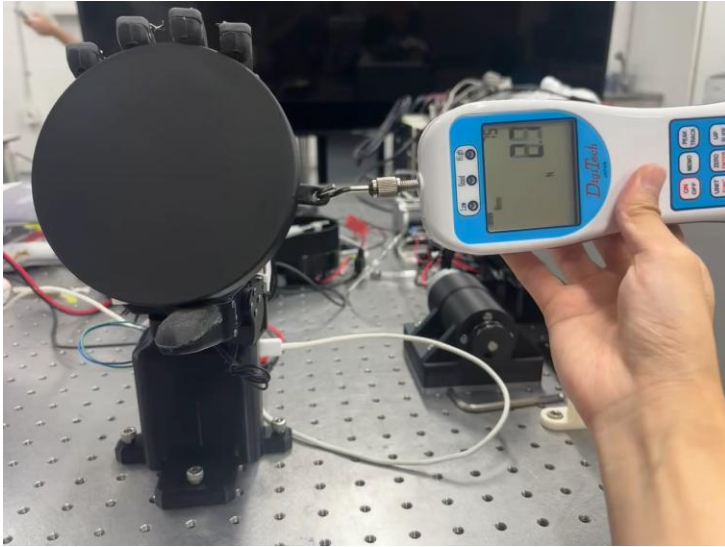
$$Score = Angle_{norm} + success_{norm}$$

Experiment – Real World

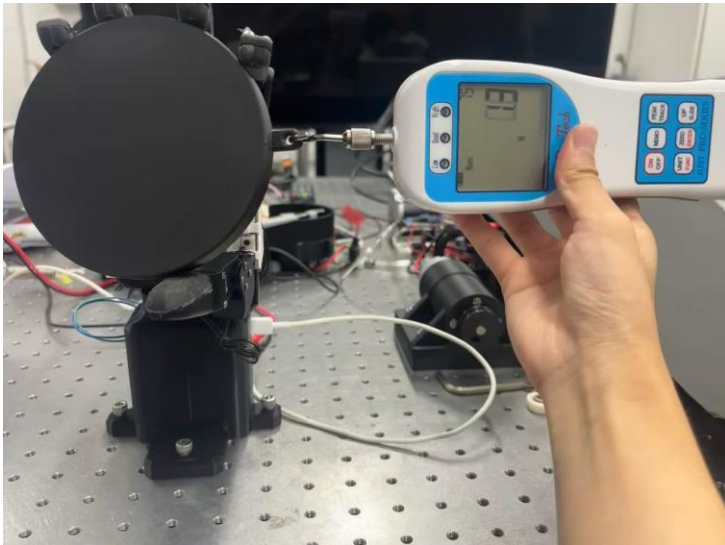


Total 10 Objects $\times$ 8 trials = 80 per k

Experiment – Real World

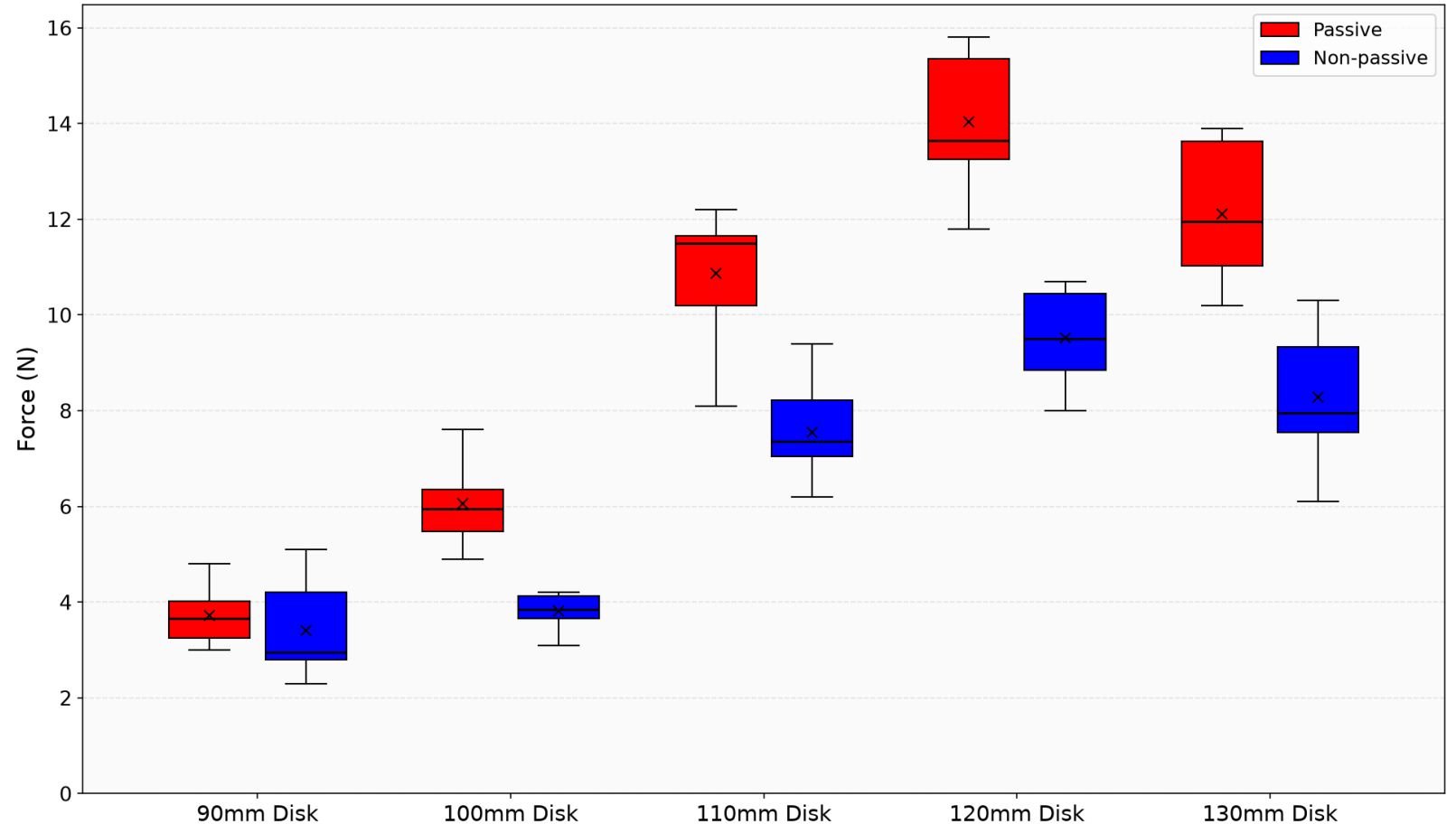


[Non passive, 130mm Disk, Max 7.6 N]



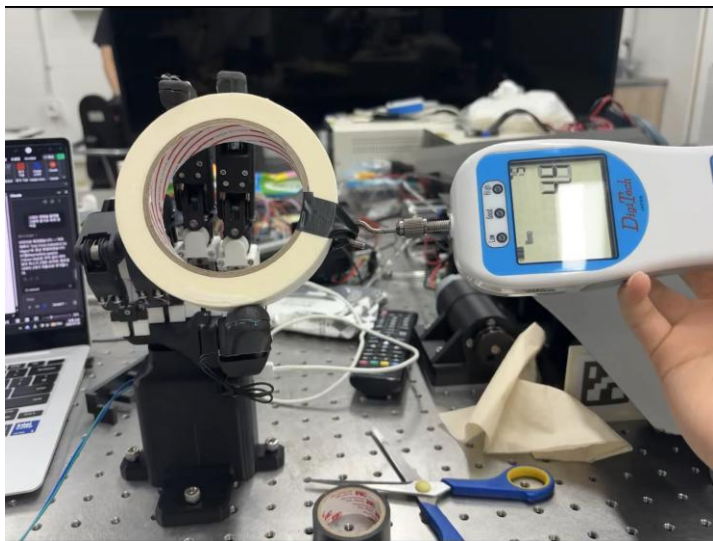
[Passive, 130mm Disk, Max 13.7 N]

Grip Force Distribution by Disk Diameter — Lateral Pull



- Starting from 90mm, a gap begins to appear and widens up to 120mm
- At 130mm, the gap stays about the same, but the absolute force starts to decline

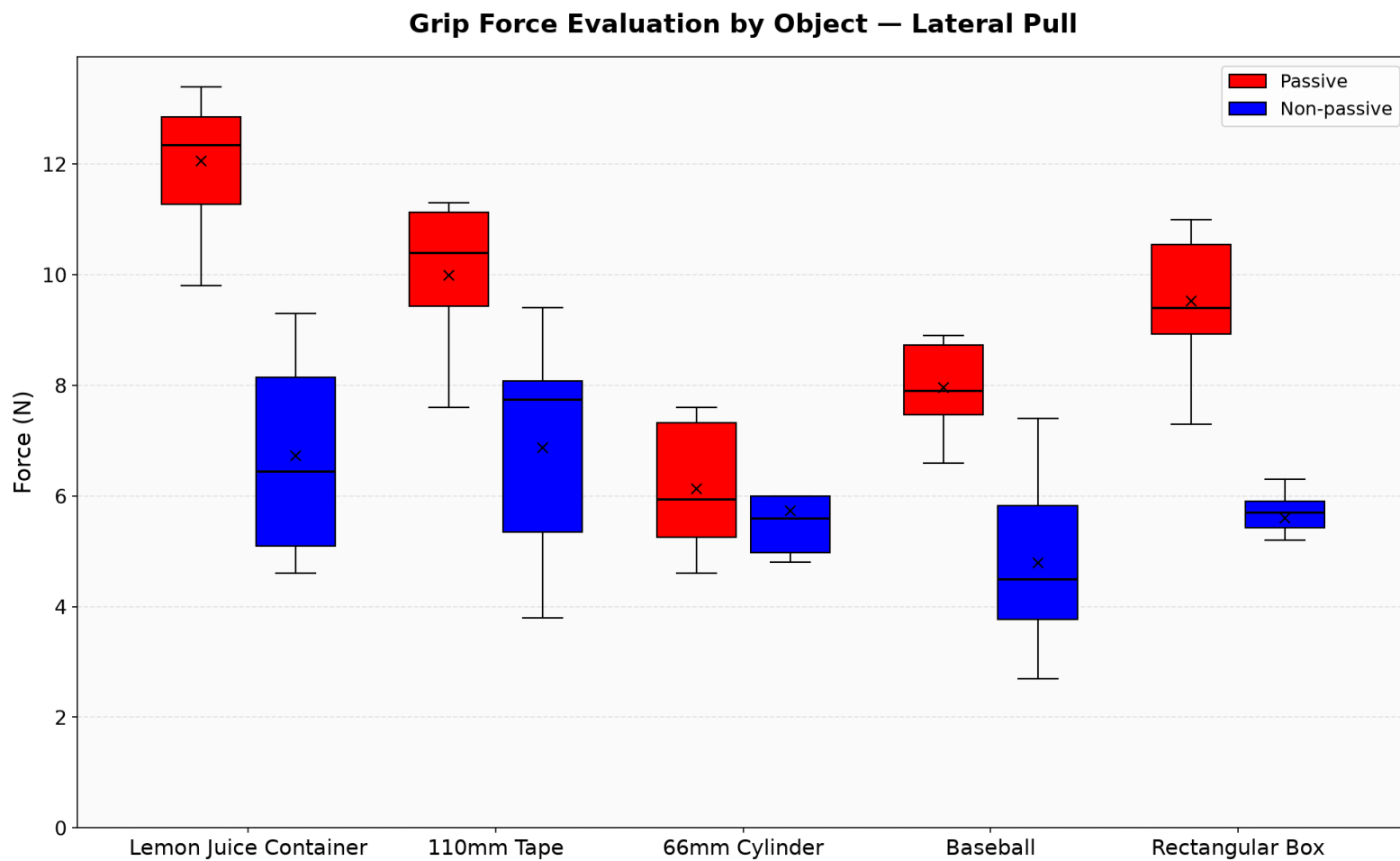
Experiment – Real World



[110mm Tape, Max 8.0 N]

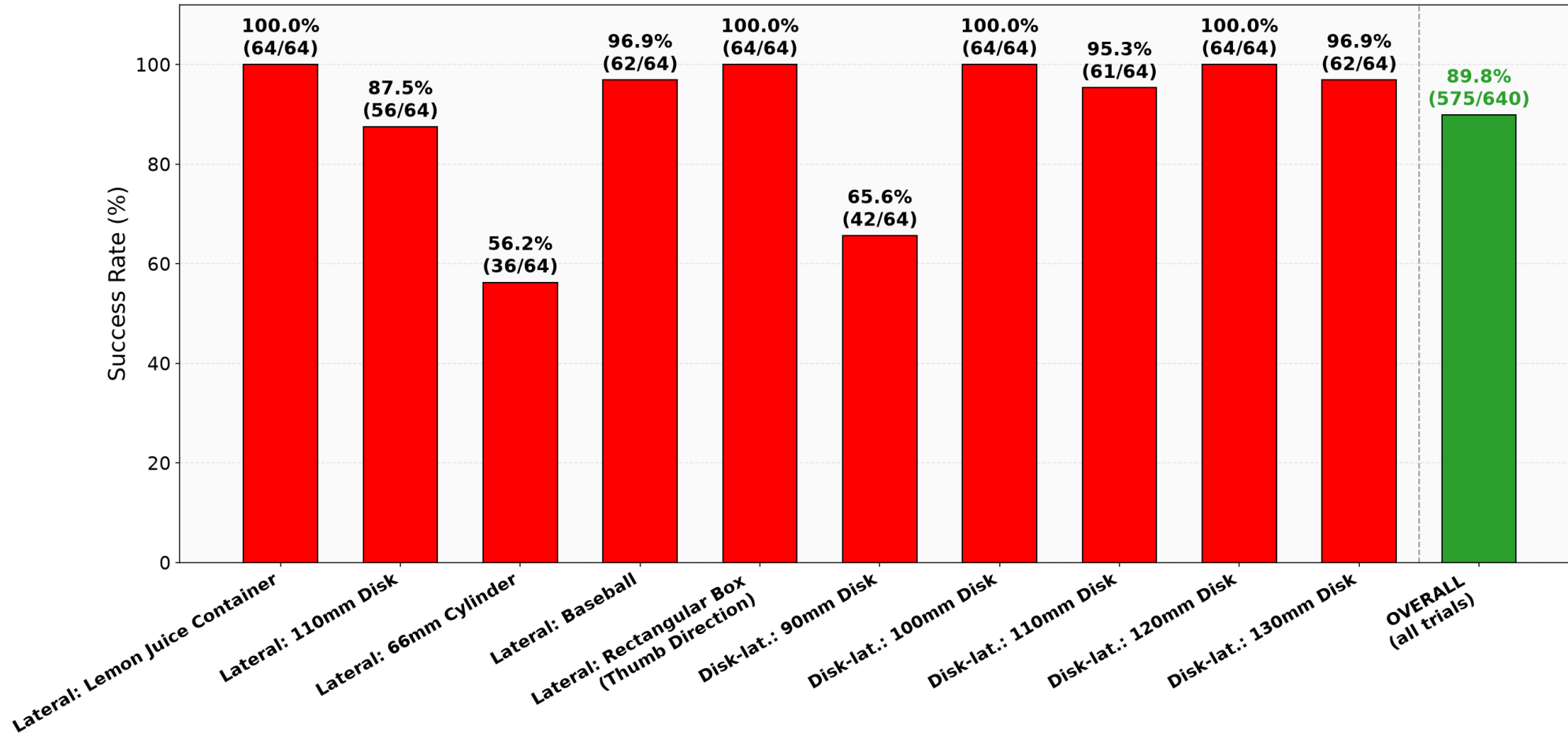


[Lemon Juice container, Max 14.3 N]



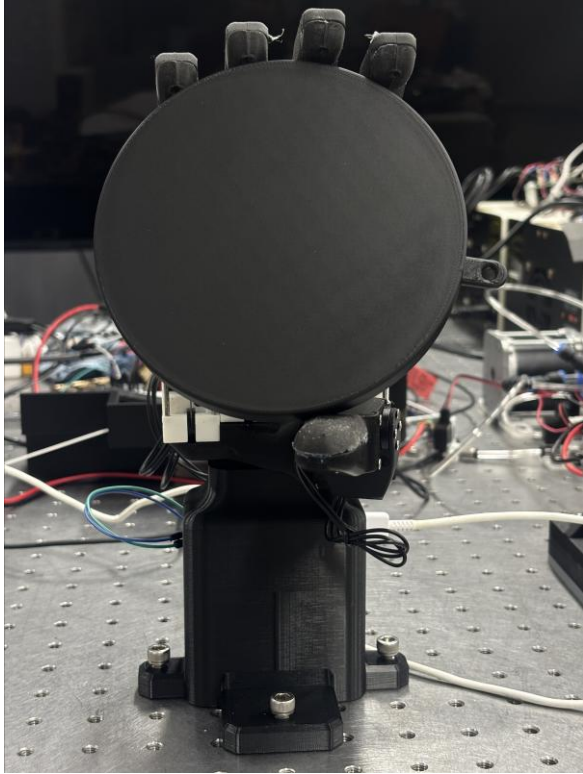
Discussion

Real Experiment — Success Rate by Condition (pairwise, Passive > Non-passive)

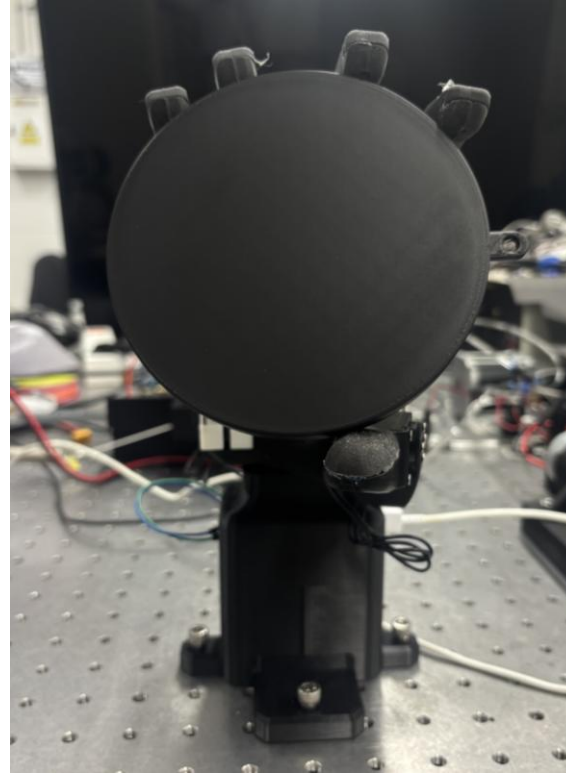


- Small diameter objects leave little room for AB/AD motion, resulting in less adaptive grasping and a lower success rate
- Real-world grasping experiments across objects achieved an 89.8% win rate

Discussion



[Non passive AB/AD Hand]



[passive AB/AD Hand]



[Adaptively maintains a stability under shaking]

- Adaptively conforms to each object's shape during grasping
- Under external disturbance, it actively re-adapts its grip to the object's shape and motion

Conclusion

- The fingers spread and adapt to the object's shape, resulting in a more stable grasp
- When the object begins to slip during pulling, the fingers additionally abduct/adduct to maintain continuous contact and provide extra supporting force
- The spring selected from simulation (86.8% success rate) was applied to the real-world environment, achieving an 89.8% success rate — validating the simulation-based selection

Future Work

- **Dual-Tendon friction**
The dual-tendon routing adds friction not fully captured in simulation
- **Modeling difficulty**
Too many bushings and pulleys make the friction both strong and hard to calculate, so it had to be simplified in simulation